

Beauty of an array

You are given a list of n positive integers. The *beauty* of an array is defined as the sum of all its elements. You have m queries represented by three integers: k , y , and x . Here's how the queries work:

1. You take any y distinct subarrays of size k from the given array and find the bitwise **AND** of their beauties.
2. Then you perform a bitwise **OR** operation on the result obtained in the first step with the integer x .
Let's say the result obtained now is z .

For each query, you need to find the maximum possible value of z after following the above described process.

Input format:

- The first line contains two integers n and m
- The second line contains n integers $A_1, A_2, \dots, A_n$

Coding question
 $1 \leq m \leq 10^3$

42min 41sec

$1 \leq A_i \leq 10^9$

$1 \leq k \leq n$

$1 \leq y \leq n - k + 1$

$1 \leq x \leq 10^9$

Output format:

The output should contain m lines, where i th ($1 \leq i \leq m$) line contains an integer denoting the answer for the i th query

Sample input ↗



Sample output

```
5 3
10 6 3 7 2
2 2 19
1 4 10
3 2 37
```

```
27
10
53
```